ITAI 2376 · Deep Learning in Artificial Intelligence
Spring 2026 | Houston City College | AI & Robotics Program

Midterm Blueprint

ZenSign AI: Zodiac-Guided Holistic Wellness Agent

Option A · Single AI Agent

1 Problem Statement

The Problem

Millions of Americans experience chronic stress, anxiety, and physical tension but cannot afford professional self-care services such as massage therapy, yoga classes, acupuncture, or relaxation coaching. A single 60-minute massage averages $75–$130; a monthly yoga membership runs $50–$150. For many working adults, single parents, and low-income households, these options are simply out of reach.

Who Has This Problem

The target user is any adult who feels physically or emotionally out of balance and cannot access paid wellness services. This includes hourly workers, students, caregivers, and anyone navigating stress without a support system or wellness budget.

Why It Matters

Untreated chronic stress contributes to cardiovascular disease, poor sleep, reduced immune function, and burnout. A free, accessible, personalized wellness agent can close the gap between need and access, delivering tailored relaxation and self-care guidance to anyone with a phone or computer, at no cost.

The Agent Solution

ZenSign AI will greet the user, collect their zodiac sign, and ask a short set of wellness check-in questions (sleep quality, stress level, energy, mood, and body tension). It will then generate a personalized, free-to-follow holistic wellness plan, including breathwork, guided meditation scripts, stretching routines, journaling prompts, and elemental self-care rituals, aligned to the user's astrological element and current wellness state.

2 Option Choice and Justification

Chosen Option: Option A Single AI Agent

ZenSign AI is designed as a single, focused agent because the entire user journey, collecting information, reasoning about astrological and wellness profiles, and generating a personalized plan can be handled cleanly within one reasoning loop. Adding separate agents would introduce unnecessary orchestration complexity for a task that is fundamentally conversational and linear.

A single agent also makes the system easier to test, debug, and explain, which matters both for this course and for real-world deployment where transparency and reliability are important to users seeking wellness support.

3 Agent Architecture and Diagram

The diagram below shows the full information flow through the ZenSign AI agent, from user input through deep learning classification to personalized plan output.

INPUT	**User Message** (zodiac sign + wellness check-in answers via chat interface)

↓ *"sends text responses for intent and sentiment analysis"*

DEEP LEARNING MODEL	**BERT Sentiment + Intent Classifier** (Module 05 Transformers) Classifies user's emotional tone and wellness intent; flags urgency signals (e.g., severe burnout, crisis language)

↓ *"sends classified intent + sentiment score to agent brain"*

AGENT BRAIN	**LLM (Claude claude-sonnet-4-20250514 via Anthropic API)** Reads zodiac element, sentiment score, and wellness inputs; reasons about which self-care practices fit the user's state; decides which knowledge base entries to retrieve

↓ *"queries wellness knowledge base by zodiac element + need"*

TOOL	**Zodiac Wellness Knowledge Base (ChromaDB vector store)** Contains curated, free self-care practices organized by astrological element (Fire, Earth, Air, Water) and wellness category (breathwork, stretching, journaling, meditation, ritual)

↓ *"returns matching practices and scripts"*

OUTPUT	**Personalized Holistic Wellness Plan** Delivered as a structured, conversational response: 5-day plan with daily breathwork, movement, journaling, and elemental self-care rituals all free, all actionable

4	**Deep Learning Connection**

Module Connection 1 Transformers (Module 05)

The agent uses a fine-tuned BERT model (Bidirectional Encoder Representations from Transformers) to classify the sentiment and intent of the user's check-in responses. BERT is the right choice here because it understands the full context of a sentence, not just individual keywords, which is critical when a user says something like "I'm okay I guess" (low energy, possibly suppressed stress) versus "I'm exhausted and overwhelmed" (high urgency). The classifier outputs a sentiment score and an intent category (e.g., stress-relief, sleep-support, energy-boost) that the LLM brain uses to personalize its reasoning.

Module Connection 2 Agentic AI and LLM Reasoning (Module 10)

The LLM (Claude) serves as the agent's reasoning engine, the AGENT BRAIN. This connects directly to Module 10 (Agentic AI), which covers how LLMs are used not just for text generation but for goal-directed reasoning, tool use, and multi-step decision-making. The LLM reads the user's zodiac element, the BERT sentiment score, and the retrieved wellness entries, then reasons about which combination of practices will best serve this specific user on this specific day. It writes the final plan in a warm, encouraging voice, which is itself a generation task.

Module Connection 3 Embeddings and Semantic Search (Module 06)

The Zodiac Wellness Knowledge Base uses dense vector embeddings (sentence transformers) to match the user's wellness needs to the most relevant self-care practices in the database. This connects to Module 06 (Embeddings and Representation Learning). Rather than keyword matching, the retrieval step uses cosine similarity between the user's need vector and the stored practice vectors ensuring that a user who says "my mind won't stop racing" retrieves breathwork and meditation content, not generic advice.

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5 Agent Framework

Chosen Framework: LangChain

LangChain is the most appropriate framework for ZenSign AI because it offers native support for the three core capabilities this agent requires: conversational memory (to maintain the multi-turn check-in dialogue), tool integration (to connect the BERT classifier and the ChromaDB knowledge base), and RAG (Retrieval-Augmented Generation) pipelines, which is exactly the pattern this agent uses to retrieve relevant wellness practices before generating the output plan.

Why not CrewAI or AutoGen?

CrewAI and AutoGen are optimized for multi-agent collaboration where distinct agents hand work to each other. Since ZenSign AI is a single agent with a linear reasoning loop, those frameworks would add orchestration overhead without any benefit. LangChain's single-agent LCEL (LangChain Expression Language) chains are a cleaner fit.

Current Status

The LangChain quickstart tutorial has been reviewed and the basic question-answering chain pattern is understood. The ChromaDB integration tutorial has also been explored. This will be the first full agent built with LangChain from scratch.

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6 Tools and Data

Tools and APIs

- ChromaDB (free, open source): Vector database for the Zodiac Wellness Knowledge Base. The agent needs this to retrieve relevant self-care practices using semantic similarity rather than keyword search, ensuring personalized and contextually relevant results.
- Hugging Face Inference API BERT Sentiment Model (free tier): Hosts the fine-tuned BERT classifier that analyzes the user's check-in responses for sentiment and intent. Using the Hugging Face API means no GPU is required locally.
- Anthropic API Claude (free credits for new accounts): Powers the LLM agent brain. Claude reads the classified intent, zodiac profile, and retrieved wellness entries and generates the final personalized wellness plan in a warm, supportive voice.
- LangChain Memory (ConversationBufferMemory, built in): Maintains the multi-turn conversation so the agent can ask check-in questions across several exchanges and remember all answers before generating the plan.

Data Sources

The Zodiac Wellness Knowledge Base will be self-generated, a curated collection of 120–200 entries covering breathwork scripts, stretching routines, journaling prompts, meditation guides, and elemental rituals, each tagged by zodiac element (Fire: Aries, Leo, Sagittarius; Earth: Taurus, Virgo, Capricorn; Air: Gemini, Libra, Aquarius; Water: Cancer, Scorpio, Pisces) and wellness category.

Because this data is self-created, there are no privacy, licensing, or ethical concerns. The content will draw on publicly available wellness literature (breathwork techniques, yoga poses, journaling frameworks) rewritten in original language for the knowledge base.

7 Build Plan Week-by-Week Timeline

Week 1	April 6–13	Set up LangChain and ChromaDB in Google Colab. Write and embed the first 50 knowledge base entries (breathwork, stretching, journaling) organized by zodiac element. Verify that a basic semantic search query retrieves relevant entries.
Week 2	April 14–20	Connect the Hugging Face BERT sentiment classifier as a LangChain tool. Build the multi-turn check-in dialogue chain using ConversationBufferMemory. Test with at least 12 simulated user inputs across different zodiac signs and stress levels.
Week 3	April 21–27	Build the full agent loop: receive zodiac input, run sentiment classification, retrieve knowledge base entries, generate personalized 5-day wellness plan. Test with 20 diverse user profiles. Refine prompts and retrieval logic based on output quality.
Week 4	April 28–May 3	Record demo video (8 min max) showing the agent handling at

		least 3 different user profiles. Write the final write-up (2–4 pages). Organize code into a clean GitHub repository with README and setup instructions. Submit all files through Canvas.

8 Anticipated Challenges

Challenge 1: BERT Classifier Accuracy on Casual Wellness Language

The fine-tuned BERT model may struggle with informal, ambiguous phrasing common in wellness check-ins. For example, "I'm fine" said by someone who is clearly not fine, or "kinda stressed I guess." Off-the-shelf sentiment models are trained on review text and may not transfer well to this domain. Mitigation: test with at least 30 diverse wellness-style sentences before integrating. If accuracy is poor, the fallback is to use a zero-shot LLM classification step inside the chain rather than a separate BERT model, which trades some speed for better contextual understanding.

Challenge 2: Knowledge Base Coverage and Retrieval Relevance

With a self-generated knowledge base of 120–200 entries, there is a risk that some user needs will not match any entry well, especially for niche zodiac sign combinations with specific wellness concerns. Retrieval may return loosely relevant results. Mitigation: implement a relevance score threshold in ChromaDB so that low-confidence retrievals trigger a fallback to a broader category search rather than returning a poor match.

Challenge 3: Responsible Framing of Wellness Advice

The agent must be carefully prompted to position itself as a supportive wellness companion, not a medical or mental health professional. There is a real risk that users in genuine distress could interpret the output as clinical advice. Mitigation: the system prompt will include explicit instructions for the LLM to add a gentle disclaimer when urgency signals are detected (e.g., "If you are experiencing a crisis, please reach out to a mental health professional or crisis line"), and to always frame suggestions as complementary self-care rather than treatment.